Genaro **Suárez**

Curriculum Vitae (as of October 30, 2026)

Current Position

Postdoctoral Fellow
Department of Astrophysics,
American Museum of Natural History
New York, NY, USA

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Homepage

Research Interests

Brown dwarf and exoplanet atmospheres, substellar variability and clouds, star and brown dwarf formation, initial mass function, scientific software development

Education

Jan. 2015–Feb. 2019 — Ensenada, Mexico
Ph.D. in Astrophysics.
Institute of Astronomy, Universidad Nacional Autónoma de México
Thesis: "Towards a Complete Study of the Initial Mass Function and Early Kinematics Evolution of the 25 Orionis Stellar Group".
Advisors: Dr. Carlos. G. Román Zúñiga, Dr. Juan J. Downes
Graduted with Honors and nominated for the "Medalla Alfonso Caso" by UNAM.
Best PhD Thesis in Astronomy in Mexico defended in 2019 or 2020 awarded by the Sociedad Mexicana de Física. PDF

Jan. 2013–Jan. 2015 — Ensenada, Mexico
M.Sc. in Astrophysics.
Institute of Astronomy, Universidad Nacional Autónoma de México
Thesis: "Towards a Complete Study of the Initial Mass Function and Early Kinematics Evolution of the 25 Orionis Stellar Group".
Advisors: Dr. Carlos. G. Román Zúñiga, Dr. Juan J. Downes
Awarded the "Premio Paris Pişmiş" by the Institute of Astronomy, UNAM and nominated for the "Medalla Alfonso Caso". PDF

Aug. 2007–Jan. 2013 — Zacatecas, Mexico
B.Sc. in Physics.
Unidad Académica de Física, Universidad Autónoma de Zacatecas
Thesis: "Spectroscopic and Photometric Study of the Ionized Regions Sh2-90 and Sh2-187".
Advisors: MSc. Marco A. Moreno Corral, Dr. Alejandro González Sánchez
Awarded the "Premio Paris Pişmiş" by the Institute of Astronomy, UNAM. PDF

Research Experience

Sep. 2022–present — New York, USA
Postdoctoral Fellow (Supervisor: Dr. Jacqueline Faherty)
Department of Astrophysics, American Museum of Natural History

Sep. 2019–Sep. 2022 — London, ON, Canada
Postdoctoral Fellow (Supervisor: Dr. Stanimir Metchev)
Department of Physics & Astronomy, The University of Western Ontario

Jan. 2015–Feb. 2019 — Ensenada, Mexico
Ph.D. Student
Institute of Astronomy, Universidad Nacional Autónoma de México

Six international research visits.

Grants and Fellowships (PDF)

Total amount of USD ≈1.5M awarded through grants and fellowships from diverse institutions (Co-I amounts reflect funding allocated to my research program), plus >USD $80k for professional activities. Selected examples:

2026 JWST Cycle 5 GO 11448 program (co-PI): USD $658,476.
2025 JWST Cycle 4 GO 7686 program (co-I): USD $25,000.
2024 JWST Cycle 3 GO 5474 program (PI): USD $316,700.
2024 JWST Cycle 2 GO 4668 program (co-I): USD $195,000.

2023 JWST Cycle 2 Survey 3930 program (co-PI): USD $119,996.

2013–2019 Mexican CONACYT Fellowships: USD $51,000.

Publications (PDF ADS ORCID Google Scholar Scopus Zotero)

Total Publications: 84 peer-reviewed journal articles (9 first author, 11 second/third author, and 64 co-author).

h**-index: 25.** Citations: 7,154 (285 first-author and 405 second/third-author).

Nine first-author articles. Selected papers:

2026 **SEDA: Spectral Energy Distribution Analyzer for Forward Modeling and Empirical Analysis of Ultracool Objects**
Genaro Suárez, Kelle Cruz, Jacqueline K. Faherty *et al.*
2026, AJ, submitted.

2025 **Diversity of Cold Worlds: Predicted Near-to-mid-infrared Spectral Signatures of a Cold Brown Dwarf with Potential Auroral Heating**
Genaro Suárez, Jacqueline K. Faherty, Ben Burningham *et al.*
2025, ApJ, 993, 165. DOI ADS

2023 **Ultracool Dwarfs Observed with the Spitzer Infrared Spectrograph: Equatorial Latitudes in L Dwarf Atmospheres are Cloudier**
Genaro Suárez, Johanna Vos, Stanimir Metchev, Jacqueline Faherty & Kelle Cruz
2023, ApJL, 954, 6S. DOI ADS

2023 **Ultracool dwarfs observed with the Spitzer Infrared Spectrograph - III. Dust grains in young L dwarf atmospheres are heavier**
Genaro Suárez and Stanimir Metchev
2023, MNRAS, 523, 4739. DOI ADS

2022 **Ultracool Dwarfs Observed with the Spitzer Infrared Spectrograph. II. Emergence and Sedimentation of Silicate Clouds in L Dwarfs, and Analysis of the Full M5–T9 Field Dwarf Spectroscopic Sample**
Genaro Suárez and Stanimir Metchev
2022, MNRAS, 513, 5701. DOI ADS

2019 **System IMF of the 25 Ori group from planetary-mass objects to high-intermediate-mass stars**
Genaro Suárez, Juan José Downes, Carlos Román-Zúñiga *et al.*
2019, MNRAS, 486, 1718. DOI ADS

Eleven second/third-author articles. Selected papers:

2026 **Silicate clouds, SiO, and molecular trends across the L-T sequence: a JWST/MIRI LRS spectral library of brown dwarfs**
Sam Barber, Stanimir Metchev, **Genaro Suárez** *et al.*
2026, MNRAS, 550, 1387. DOI ADS

2026 **Inspecting Cloudy Substellar Atmospheres with JWST MIRI Synthetic Magnitudes from Spitzer Mid-infrared Spectra**
Jolie L'Heureux, **Genaro Suárez** *et al.*
2026, ApJ, 1003, 236. DOI ADS

2026 **Clouds with a silicate lining: Using JWST spectra to probe atmospheric diversity in young AB Dor L dwarfs**
M. B. Lam, J. M. Vos, **G. Suárez** *et al.*
2026, A&A, 709, 56. DOI ADS

2026 **The stellar initial mass function of nearby young moving groups**
R. Bertolotto-Stefanelli, J. J. Downes, **G. Suárez** *et al.*
2026, A&A, 706, 370. DOI ADS

Sixty co-author articles. Selected collaborations:

2026 **The Diversity of Cold Worlds: Age and Characterization of the Exoplanet COCONUTS-2 b**
R. Kiman, C. A. Beichman, A. Ruiz Diaz, J. Faherty, B. Lacy, **G. Suárez** *et al.*
2026, AJ, 171, 60. DOI ADS

2025 **Detection of the Silicate Precursor Silane in a Cold, Low-Metallicity Brown Dwarf**
J. Faherty, A. M. Meisner, B. Burningham, C. Visscher, M. Line, **G. Suárez** *et al.*
2025, Nature, 628, 511. DOI ADS

2024 **Methane emission from a cool brown dwarf**
J. Faherty, B. Burningham, J. Gagné, **G. Suárez** *et al.*
2024, Nature, 628, 511. DOI ADS

Five first-author conference proceeding.

Awards and Honors

2026 **Nominee** for the Blavatnik Regional Awards for Young Scientists.

2021 **Best PhD Thesis in Astronomy in Mexico (2019 and 2020).** PDF
By the Sociedad Mexicana de Física/División de Astrofísica.

2019 and 2015 **UNAM grant.**
For on-time completion of Master's and Ph.D. programs at UNAM.

2019 and 2015 **Nominee** for the Medalla Alfonso Caso by UNAM. PDF
For exceptional master's and PhD thesis defenses.

2016 and 2014 **Premio Paris Pişmiş.** PDF PDF
Excellence Award for undergraduate/graduate students by the Institute of Astronomy, UNAM.

Research Talks and Posters (PDF) (PDF)

Fifty-one research talks (26 invited presentations) at national and international conferences and seminars. Selected talks:

Jun. 25, 2026 Shanghai, China **Invited seminar** "Understanding Substellar Atmospheres Through Silicate Clouds", Tsung-Dao Lee Institute.

Jun. 18, 2026 Tokyo, Japan **Contributed talk** "Silicate Clouds as Drivers of Spectral Diversity in Ultracool Atmospheres", Cool Stars 23, TOC Ariake.

May. 19, 2026 Baltimore, MD, USA **Invited seminar** "Understanding Substellar Atmospheres Through Silicate Clouds", Space Telescope Science Institute.

Jun. 07, 2026 Phoenix, AZ, US **Invited talk** "Predicted Near- to Mid-infrared Spectral Signatures of a Cold Brown Dwarf with Potential Auroral Heating", 247th AAS Meeting, Phoenix Convention Center.

Jun. 26, 2025 Cork, Ireland **Invited review talk** "What Observations Have Taught us About 4D Extrasolar Atmospheres?" at the 225 European Astronomical Society Annual Meeting.

Jul. 01, 2024 Pasadena, CA, USA **Invited talk** "Explaining the Diversity of Cold Exoworlds with JWST", California Institute of Technology.

Jun. 28, 2024 San Diego, USA **Contributed talk** "Explaining the Diversity of Cold Exoworlds with JWST" at Cool Stars 22.

Jan. 9, 2024 New Orleans, USA **Press conference** "Extrasolar Worlds Exhibit Exotic Sandy Clouds at the Equator" at the 243rd AAS meeting.

Participation in 29 conferences, 7 congresses, 11 summer schools, and 9 workshops.

Eleven first-author posters in conferences. Selected posters:

Jul. 7–11, 2025 Montreal, Canada
"Cloudy Extrasolar Skies in the JWST Era" at the Exoclimes VII conference.

Dec. 12–15, 2022 Baltimore, MD, USA
"Gravity-dependent Sedimentation of Dust Clouds in Ultra-cool Atmospheres" at the First Science Results from JWST Conference.

Awarded Telescope Time and Observing Experience (PDF)

Observing programs across multiple facilities, including 21 accepted JWST programs (1 PI, 2 Co-PI), and others using facilities such as the NASA Infrared Telescope Facility, Gran Telescopio Canarias, and Gemini Observatory.

Telescope time awarded as PI:

- 57 hr with IRTF/SpeX
- 3 night with MMT/Hectospec
- 1 hr with CT-4m/DECam
- 7 hr with GTC/OSIRIS
- 10 hr with SDSS-IV/APOGEE-2
- 12 nights with OAN-SPM/Echelle
- **20 hr with JWST MIRI as PI (Cycle 3 GO 5474).**
- **113 hr with JWST MIRI as co-PI (Cycle 5 GO 11448).**
- **Survey program with JWST MIRI as co-PI (Cycle 2 Survey 3930).**

Academic and Community Service (PDF)

Panel reviewer for NASA (1), referee for academic journals (13), member of graduate thesis committees (1), reviewer for telescope time allocations (4), SOC member (4) and session chair (5) at conferences, and additional departmental and community service. Selected examples:

2026 **Referee** for Nature and Nature Astronomy (2 papers).

2026 **Panel Reviewer** for NASA Exoplanets Research Program (XRP).

2022–2026 **Referee** for the American Astronomical Society (AAS) (6 papers).

2024–2026 **Session Chair** at the 243rd, 245th, and 247th AAS meetings.

2022–2026 **Referee** for Astronomy & Astrophysics (4 papers).

2023 and 2024 **External Expert Reviewer** for JWST Cycle 3 and Cycle 4.

Supervision and Mentorship Experience (PDF)

Supervisor/co-supervisor of 4 graduate, 11 undergraduate, and 14 high-school students. Selected examples:

2026–present **Supervisor** of master's students Alejandra Ross and Gerally Medina Rivera at CUNY.

2026–present **Supervisor** of undergraduate student Luis Diego Lopez Perez, Universidad de San Carlos, Guatemala, through the CENCA BRIDGE program.

2022–present **Supervisor** of 13 high-school students in the Science in Research Mentoring Program at AMNH.

2024–2026 **Supervisor** of 4 REU students at AMNH.

2023–2024 **Supervisor** of master's student Alexandra Baldelli, Columbia University (*now Ph.D student at University of Delaware*).

2023–2024 **Supervisor** of undergraduate student Jolie L'Heureux, Columbia University (*now Ph.D student at CUNY*).

2022–2024 **Co-supervisor** of master's student Rafael Bertolotto, Universidad de la República, Uruguay (*now Ph.D student at University of Arizona*).

2022–2024 **Co-supervisor** of undergraduate student Sherelyn Alejandro Marchán, CUNY (*now Ph.D student at CUNY*).

Teaching Experience (PDF)

Lectured graduate-level courses on Exoplanets, Brown Dwarfs, and the Initial Mass Function at different institutions, and served as instructor for workshops on data analysis, programming, and stellar astrophysics. Selected examples:

Jul. 2025, Jul. 2026 Dublin, Ireland Brownsville, TX; San Francisco, CA, USA — **Instructor** for workshops on modeling observations and deriving fundamental properties of extrasolar worlds using the SEDA code at Trinity College Dublin, The University of Texas Rio Grande Valley, and San Francisco State University.

Mar. 2025 New York, NY, USA — **Lecturer** on brown dwarf and gas giant planet atmospheres, formation, and evolution for graduate students at Columbia University.

Fall 2024 New York, NY, USA — **Instructor** for the "Introductory Computer Programming" course in the Science Honor Program for high school students at Columbia University.

Summer 2024, 2025 New York, NY, USA — **Instructor** for nine MAT residents at the American Museum of Natural History.

Outreach and Engagement (PDF)

Participation in 80 outreach, community engagement, and media initiatives. Selected activities:

2022–present New York, NY, USA — **Mentor** of citizen scientists in the NASA's Backyard Worlds: Planet 9 project.

2026 New York, NY, US — **Leader and organizer** of the project Stars Beyond Barriers to support students from underrepresented groups to pursue STEM.

Nov. 11, 2025 Virtual — **Featured speaker** on the episode "¿Cómo se forman los planetas y estrellas?" for Cumbre Kids through "Cráneo: Ciencia para niños curiosos".

Summer 2025 New York, NY, USA — **Spanish translator** for the Hayden Planetarium show "Encounters in the Milky Way at the American Museum of Natural History".

2023–2025 New York, NY, USA — **Facilitator** of DEI in STEM training sessions for a total of ∼150 high school students at AMNH.

Aug. 05–06, 2024 Zacatecas, Mexico — **Leader and organizer** of the project "Bringing Astronomy Closer to Indigenous Women in Rural Communities in Mexico" as part of WGAP by NA-ROAD.

Apr. 2024 New York, NY, USA — **Volunteer speaker** on astronomy-related topics at the Columbia University's Astronomy in Prisons Program for ≈20 women at MDC Brooklyn.

2023–2024 New York, NY, USA — **Translator** of the Machine Learning for Physics and Astronomy course into Spanish at CCA Flatiron Institute.

Memberships

Active Memberships

2022–present — Member of 21 JWST Cycle 1–5 programs

2022–present — Member of the NASA's Backyard Worlds: Planet 9 citizen science project

2022–present — Member of the American Astronomical Society (AAS)

Technical Skills

Projects

SEDA — Developer of the Spectral Energy Distribution Analyzer (SEDA) python package for forward modeling and empirical analyses of observations for ultracool objects.

Computer Skills

High — PYTHON, Jupyter Notebooks, Git, Github, IDL, IRAF, DS9, TOPCAT, STILTS, VOSA, Vizier, Overleaf, LaTeX, Linux, UNIX commands, Vi/Vim, JWST pipeline

Languages

Spanish **Mother tongue**

English **Fluent**

Contact Information of Professional References

Dr. Jacqueline Faherty
Associate Curator
Department of Astrophysics
American Museum of Natural History
Phone number: +1 201 694 0807
E-mail: jfaherty@amnh.org
Current supervisor and active collaborator since 2022

Dr. Stanimir Metchev
Professor (Physics and Astronomy)
Department of Physics & Astronomy,
The University of Western Ontario
Phone number: +1 519 661 2111, 88438
E-mail: smetchev@uwo.ca
Previous postdoc supervisor and active collaborator since 2019

Dr. Johanna Vos
Associate Professor in Astrophysics
Trinity College Dublin,
The University of Dublin
Phone number: +353 83 379 5056
E-mail: Johanna.Vos@tcd.ie
Close and active collaborator since 2022

Dr. Kelle Cruz
Associate Professor
Department of Physics and Astronomy,
Hunter College, City University of New York
Phone number: +1 917 837 9748
E-mail: kelle.cruz@hunter.cuny.edu
Close and active collaborator since 2022

Dr. Juan J. Downes
Associate Professor
Department of Astronomy
Universidad de la República
Phone number: +598 25 258 624
E-mail: jdownes@fcien.edu.uy
Former PhD advisor and active collaborator

Dr. Adam Burgasser
Professor
Department of Astronomy and Astrophysics
University of California San Diego
Phone number:
E-mail: aburgasser@ucsd.edu
Active collaborator since 2023